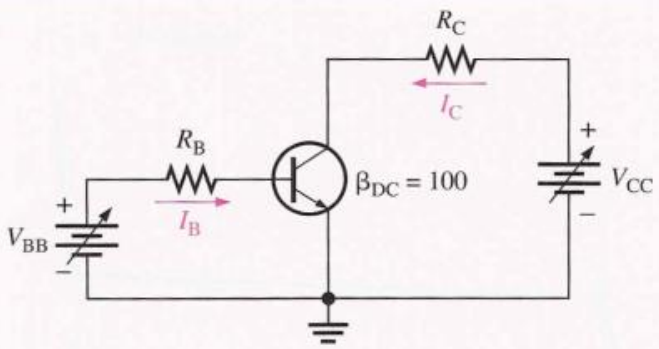
Tutorial 2.1

Problem 1: Sketch collector curves for the following circuit for I_b = 5 µA to 25 µA with 5µA increments. Assume β = 100 and V_{CE} does not exceed breakdown.

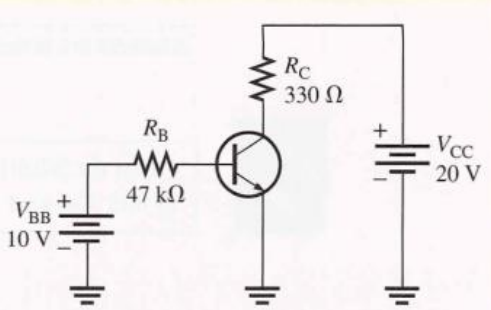
FIGURE 4–10



Problem 2:

Determine the Q-point for the circuit in Figure 5–7. Find the maximum peak value of base current for linear operation. Assume $\beta_{DC} = 200$.

FIGURE 5–7



Problem 3:

EXAMPLE 4.3 Given the load line of Fig. 4.16 and the defined *Q*-point, determine the required values of V_{CC}, R_C, and R_B for a fixed-bias configuration.

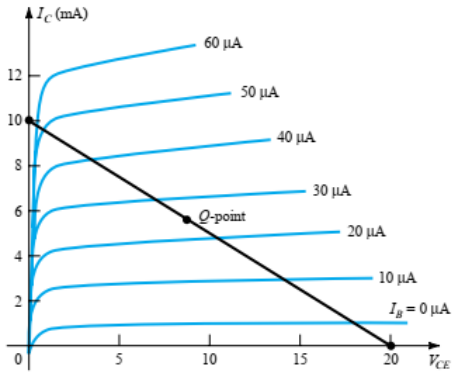
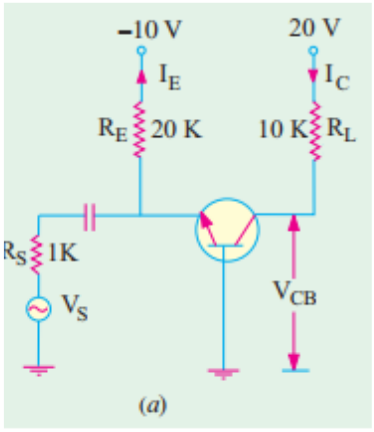


Figure 4.16 Example 4.3

Problem 4:

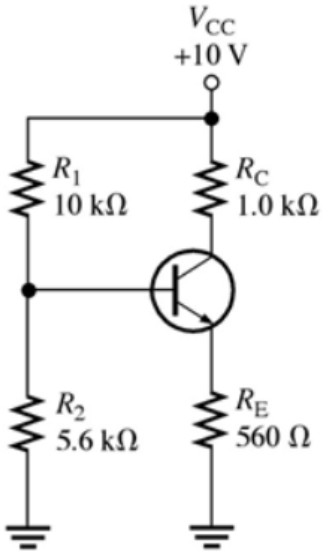
Example 58.2. *In the CB circuit of Fig. 8.3 (a), find*
(a) dc operating point and dc load line
(b) maximum peak-to-peak unclipped signal



Tutorial 2.2

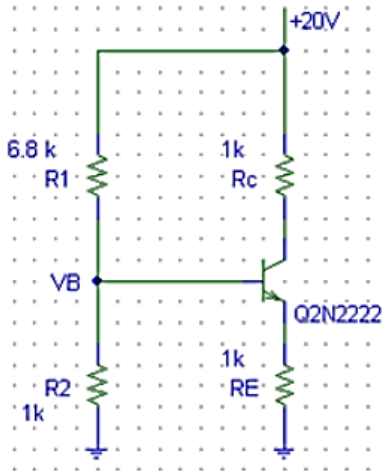
Problem 1:

Example 8: Determine V_{CE} and I_C in the voltage-divider biased transistor circuit. Assume $\beta_{DC} = 100$ and $I_E \cong I_C$.



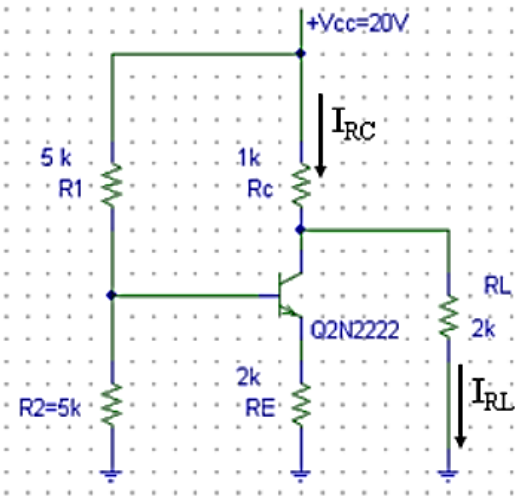
Problem 2:

Example 9: Determine V_{CE} and I_C in the voltage-divider biased transistor circuit. Assume $\beta_{DC} = 50$ and $I_E = I_B + I_C$.

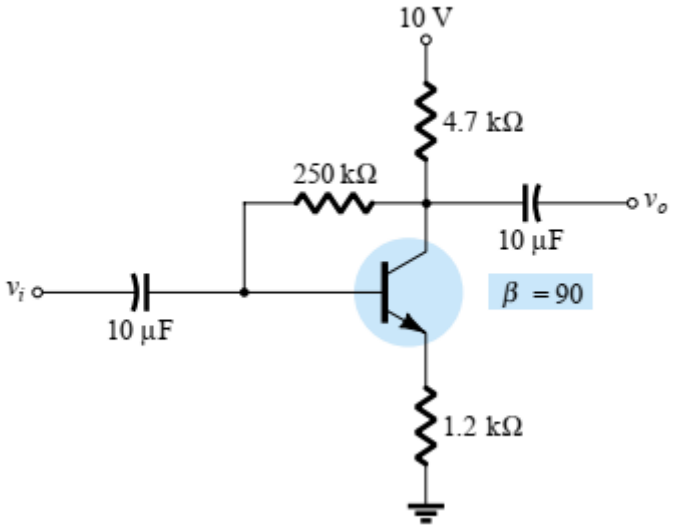


Problem 3:

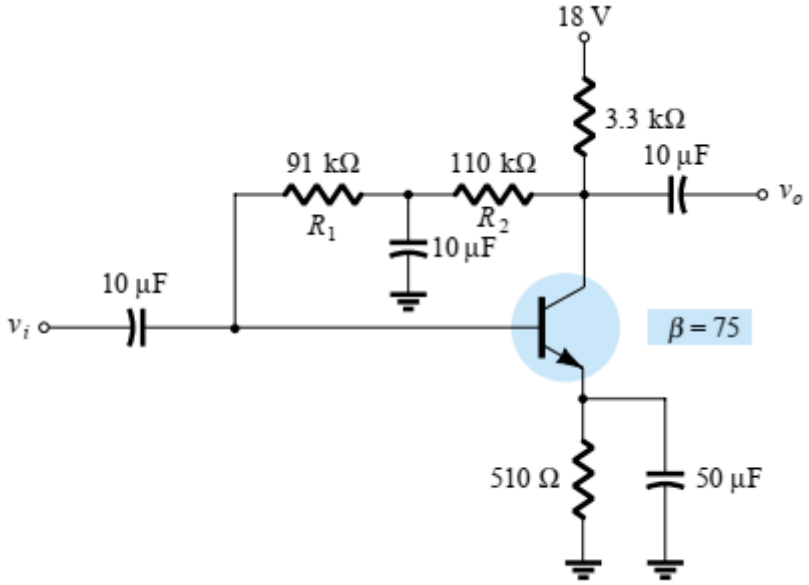
Example 10: Determine Q-point (I_C,V_{CE}), I_{RC} and I_{RL}. Assume $\beta_{DC} = 200$ and $I_E \cong I_C$.



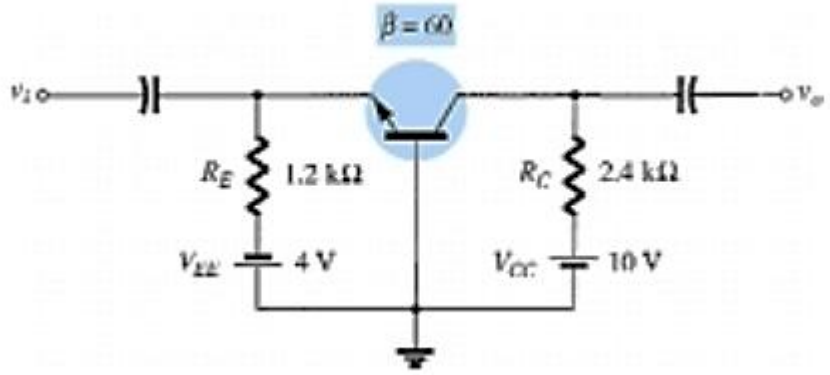
Problem 4: Draw the loadline for the circuit:



Problem 5: Determine the base current Ib and collector voltage Vc of the following circuit:



Problem 6: Determine the Vcb and Ib for the circuit:



Problem 7: Complete the circuit values based on the information given:

